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Microwave assisted alkaline deep eutectic solvents treatment inducing deconstruction of recalcitrant moso bamboo cell walls for highly efficient dual-enzymatic saccharification

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ABSTRACT

Green and efficient deconstruction and valorization of lignocellulosic biomass remain as a hot topic for biorefinery research. Herein, novel alkaline arginine-based deep eutectic solvents (DESs) pretreatment was employed to achieve efficient fractionation of highly recalcitrant moso bamboo. Higher solvent basicity was revealed for the proposed DESs with Kamlet-Taft solvatochromic parameters values of α (0.292), β (0.845), and π* (1.284). Thus, a high (β-α) value was achieved, favorable for forming stronger hydrogen bonds with lignin and xylan for subsequent separation. Supramolecular structural changes of DESs-treated bamboo cellulose were quantitatively calculated, suggesting increased crystallinity index and decreased crystallite size that were beneficial for the following enzymatic degradation. The DESs pretreatment dramatically increased the removal rate of lignin (as high as 94.4 %) and xylan (75.6 %), greatly enhancing subsequent enzymatic hydrolysis with the highest saccharification yield rate of 91.8 %, 9.8 folds as compared to that of raw bamboo biomass. Critical factors affecting enzymatic conversion performance of the bamboo cellulose were integrated, showing strong correlations with xylan removal ($R^2 = 0.91$), lignin removal ($R^2 = 0.99$), and cellulose crystallite size reduction ($R^2 = 0.852$). Mass balance of the present bioconversion process was performed. Based on a 100 g bamboo material balance, 51.97 g of high output fermentable sugars (including 43.96 g of glucose and 7.83 g of xylose) can be obtained after a 15-minute short pretreatment at 130 °C. Therefore, the proposed arginine-based alkaline DESs pretreatment offers new insights into the enhanced processing and bioconversion of lignocellulosic biomass.

1. Introduction

Due to overconsumption of fossil resources and serious environmental pollution, exploration for environmentally friendly and renewable alternatives has received more and more attention [1]. In recent decades, biorefining of lignocellulosic biomass has been considered as an ideal approach [2,3]. However, the recalcitrance of lignocellulosic biomass, such as moso bamboo, induced by the factors of lignin content, cell wall thickness, cellulose crystallinity, polymer orientation, lignin-carbohydrate complex (LCC) linkages, hemicellulose branching, and intricate anatomical structures etc., has consistently posed a significant barrier to the efficient conversion of biomass into renewable energy and chemicals [4,5]. Therefore, exploring fascinating solvents for effective pretreatment and deconstruction of recalcitrant lignocellulose cell walls, as well as separating the major components for further bioconversion remain a hot topic but not well addressed.

Among the reported lignocellulosic pretreatment approaches, ionic liquids (ILs) have emerged as promising solvents for biomass pretreatment due to their wide liquid range, high solvating ability, and high ionic conductivity [6]. While, growing environmental and economic concerns limit their industrial applications [7]. With the introduction and development of green chemistry, the adoption of low-environmental impact processing technologies for producing high-value-added products from lignocellulosic raw materials has become a

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highly challenging task [8–10]. Recently, DESs have stood out in green biocatalysis and biotransformation with the advantages of low cost, simple synthesis, and environmental sustainability. DESs are consist of hydrogen bond acceptors (HBAs) and hydrogen bond donors (HBDs), usually using quaternary ammonium salts as HBAs, and acids, bases, carbohydrates, and phenolic substances as HBDs [11,12]. Therefore, DESs consisting different molar ratios of HBAs and HBDs may have many possible combinations, affecting their physical and chemical properties [13]. In recent years, numerous studies have been carried out using acidic DESs, such as Brønsted acidic DESs (BADESs) or Lewis acidic DESs (LADESs) for the fractionation and valorization of biomass [14]. In Lewis acidic DESs, these compounds with Lewis acidity are usually combined with other compounds (e.g., urea, ethylene glycol, etc.) to form eutectic mixtures through hydrogen bonding [15,16]. For example, the introduction of Lewis acids ($AlCl_3$, $FeCl_3$, $FeCl_2$, $ZnCl_2$ and $CuCl_2$) into choline chloride/glycerol to make a ternary DES for the efficient pretreatment of hybrid beteljack energy grasses produces high purity and oxidative-resistant lignin [17]. In Brønsted acidic DESs, these acidic compounds usually combine with hydrogen bonding acceptors (e.g., quaternary ammonium salts, alcohols, etc.) to form eutectic mixtures with low melting points and high stability [18]. For example, DESs combined with choline chloride (ChCl) and natural p-coumaric acid (PCA) was applied for pretreatment on switchgrass under 160 °C, which induced 70 % of xylan removal [19]. However, acidic DESs will provide protons to catalyze the random cleavage of glycosidic bonds in xylan, leading to its hydrolysis to xylose, which was possibly dehydrated to furfural catalyzed by Brønsted and Lewis acids [20]. These degradation products will inhibit downstream hydrolysis and fermentation, which must be achieved after the detoxification process continued efficient biotransformation [21]. Therefore, in order to maximize product revenues from lignocellulosic, it is important to adopt efficient DESs to deal with the challenges of acidic conditions. Compared to neutral or acidic environments, polysaccharides showed less degradation under mild alkaline DESs conditions, which also effectively reduce the formation of inhibitors [22]. Meanwhile, the recovered hemicellulose has a higher molecular weight, thus adding additional high economic value to the production process [23]. It is also deserved to note that, lignin condensation also occurs in the intense acid treatment, hindering further bioconversion of the treated lignocellulose. Comparatively, alkali treatment process usually does not require high temperatures and also reduces the occurrence of condensation reactions [24,25]. For example, alkaline DESs (choline chloride/glycol/Na_2S) incorporated by Na_2S greatly facilitated the decomposition of lignin with a removal rate of up to 74.67 %, while maximizing the retention of glucan and hemicellulose [26]. Thus, it is crucial and emerging to find suitable alkaline DESs to achieve efficient biotransformation.

Among the chemicals consisting of alkaline DESs, choline hydroxide (ChOH) is an aqueous, low-cost, biodegradable basic ionic liquid, which can act as effictive HBA and provide an alkaline environment for DESs [27]. For example, choline hydroxide/urea (ChOH/Ur) DESs with 11 wt % water were used to dissolve at least 9.5 wt% cellulose with the degree of polymerization (DP) of 926 within 30 min [10]. Pan and colleagues applied two choline hydroxide based DESs (choline hydroxide-glycerol and choline hydroxide-urea) pretreatment on wheat straw. It removed 73.9 % and 75.6 % of lignin respectively, which showed a better coupling effect than choline chloride acid DESs [28]. These literatures indicate that, ChOH-based DESs have great potential for pretreatment of biomass. Besides, natural origin of the components gives DESs a great edge over DESs because they are clearly less toxic and more environmentally friendly. Natural deep eutectic solvents (NADESs) include amino acid-glycosyl DESs, amino acid-organic acid-DESs, and glycosyl DESs etc. [29–31]. Recently, our group reported a novel arginine/ lactic acid DES (1:7) pretreatment strategy on moso bamboo, reaching high lignin removal rate of 84.4 % and the highest enzymatic conversion yield of 81.9 % glucose [32]. Pretreatment on corncob by DESs consisting of amino acids (lysine, arginine, histidine) and betaine for 5 h at 60 °C, about 50 % of xylan and 60 % of lignin could be removed, as well as obtaining enzymatic hydrolysis yield of 60.8 % [33]. These results demonstrated satisfactory lignin and hemicellulose extraction, along with a high hydrolysis rate, using NADESs pretreatments. The remarkable bioconversion performance makes NADESs well suited for the pretreatment of lignocellulosic biomass. However, up till now, there are very limited reports on ChOH as a prerequisite for NADESs with the cooperation of bio-derived amino acid. Moreover, the attempt to apply ChOH-based NADESs on biomass treatment and the inner mechanisms on lignocellulose deconstruction have been rarely discussed.

In this study, natural choline hydroxide/arginine (CA) DESs were developed for the first time to pretreat highly recalcitrant moso bamboo with the assistance of microwave radiation in short treatment time. The polarity and viscosity of DESs with different molar ratios were studied and their pretreatment properties were characterized. The effect of DESs on the cell wall structure of bamboo was described through scanning electron microscopy (SEM) and laser confocal technology. X-ray diffraction was used to study the variations of aggregated and crystalline cellulose under the treatment of DESs and characterization of extracted lignin by 2D HSQC NMR spectroscopy. The correlation between lignin removal, xylan removal, crystal size of cellulose, and enzymatic hydrolysis yield was analyzed to explore the influencing factors of enzymatic hydrolysis. The DESs pretreatment strategy of bamboo was evaluated by material balance accounting. The present study proposing physicochemical properties and pretreatment effects of DESs may provide a reliable basis for further application of biorefinery on lignocellulosic biomass by alkaline NADESs.

2. Materials and methods

2.1. Materials and chemicals

Moso bamboo culms were provided by Yunnan Forestry Center. Bamboo powder (40–80 mesh) was extracted by soxhlet extraction (acetone-ethanol volume ratio 2:1) for 6 h to remove pigments and alcohols. Bamboo strips were cut into 10 μm-sized transverse and radial sections using a slicer (HM 325, Thermo Scientific, USA) and observed under microscope. L-arginine and choline hydroxide were obtained from Macklin Biochemical Co., Ltd (Shanghai, China). All other chemicals were purchased from Shanghai Chemical Reagent Co., Ltd (Shanghai, China) and used as received.

2.2. DES preparation and characterization

Choline hydroxide solution was slowly dripped into arginine solution at a molar ratio of 1:1/ 2:1 with homogeneous stirring for 24 h. The mixture was rotary evaporated under negative pressure at 50 °C to remove water and dried under vacuum at 70 °C for 48 h. The prepared DESs were left overnight to observe their stability. The DES with molar ratios of 1:1 and 1:2 (arginine to choline) are labeled as CA1 and CA2, respectively, as shown in Table S1. The α, β, and π* of the prepared DESs were determined using N, N-diethyl-4-nitroaniline (NEt2), 4-nitroaniline (NH2), and Nile red (NR) as dye indicators of solvent properties [34]. The maximum absorption wavelength λmax of the dye-DES mixture was determined between 350–700 nm using a UV–Vis spectrophotometer and measured three times to minimize experimental errors. The parameters were calculated according to the following equation [35].

$$\nu_{max} = \frac{1}{\lambda_{max}} \times 10000 \quad (1)$$

$$\pi^* = 0.314 \times \left(27.52 - \nu_{max,NEt2}\right) \quad (2)$$

$$\beta = 11.134 - \frac{3580}{\lambda_{max,NH2}} - 1.125\pi^* \quad (3)$$

$$\alpha = \frac{19.9657 - 1.1024\pi^{*} - \nu_{max,NR}}{1.6078} \quad (4)$$

where $\lambda_{max,NEt2}$, $\lambda_{max,NH2}$ and $\lambda_{max,NR}$ represent the maximum absorption wavelength of N, N-diethyl-4-nitroaniline, 4-nitroaniline and Nile Red, respectively. The parameter α represents the hydrogen-bond-donating ability (i.e., Lewis acidity), β represents the hydrogen-bond-accepting ability (i.e., Lewis basicity), and π* represents the polarizability of the solvents.

2.3. DES pretreatment

Moso bamboo powder was mixed with DES at a solid–liquid ratio of 1:20 and added to PTFE microwave reactor. The mixture was heated at 600 W for 10 min and kept for 5 min using MDS-8G microwave disintegrator (SINEO, Shanghai, China). The microwave temperatures were set to 90 °C, 100 °C, 110 °C, 120 °C and 130 °C, respectively. At the end of the reaction, the solid residue was washed with 50 % acetone solution and 1 % NaOH aqueous solution, and then the solid residue was washed repeatedly with deionized water for neutrality. Part of the neutral material will be air-dried for subsequent testing and part will be placed at 4 °C for cellulase hydrolysis. The pretreated samples were named according to the type of DES and the pretreatment temperature, e.g., CA1 bamboo chips pretreated at 90 °C supplemented with microwave were called CA1-90, and the other samples were followed similarly.

The slices were fixed with nickel mesh and pretreated in DES at 90 °C for 4 h. The slices were washed with 50 % acetone solution, 1 % NaOH solution, and deionized water until neutral, then air-dried and used for subsequent microscopic examination. The CA1- and CA2-treated moso bamboo sections were respectively named CA1-90-S and CA2-90-S, as shown in Table S1.

2.4. Enzymatic hydrolysis

The enzyme activity of cellulase (Celic CTec2) used in the reaction was 250 FPU/mL. The 20 mL sample suspension with a concentration of 5 % was prepared in a 50 mL conical flask using 0.1 M citrate buffer, 60 μL of tetracycline, and 20 FPU/g dextran cellulase. The reaction system was subjected to enzymatic hydrolysis in a water bath at 50 °C for 72 h. The supernatant was periodically taken to inactivate and centrifuged. The monosaccharides in the supernatant were analyzed by Shimadzu high-performance liquid chromatography. All the enzymatic hydrolysis were conducted at least triplicates and average results were used.

$$Conversionyield(\%) = \frac{Releasedglucoseamount(g)}{Initialamountofglucaninsolids(g)} \times 100\%\# \quad (5)$$

2.5. Characterization

2.5.1. Chemical compositional analysis

The chemical composition of the samples was analyzed by the National Renewable Energy Laboratory (NREL) [36]. Glucose and xylose obtained after sulphuric acid hydrolysis were quantified by liquid chromatography.

$$Solidrecovery(\%) = \frac{Recoveredsolidresidue(g)}{Rawbiomass(g)} \times 100\% \quad (6)$$

$$Celluloserecovery(\%) = solidrecovery(\%) \times \frac{celluloseinsolidresidue(g)}{celluloseinrawbiomass(g)}\# \quad (7)$$

$$Lignin/xylanremoval(\%) = 1 - \left(solidrecovery(\%) \times \frac{lignin/xylaninsolidresidue(g)}{lignin/xylaninrawbiomass(g)}\right)\#\#\#\#\#\# \quad (8)$$

2.5.2. Crystal structural analysis

X-ray diffractometer n (XRD) (Ultima IV, Rigaku, Japan) was used for the determination of bamboo powder. The scanning range was 2θ = 5°–50°, the scanning speed was 5°/min, the scanning step was 0.02°, the wavelength λ = 1.5419 Å, the voltage was 40 kV, and the current was 40 mA. The cellulose crystal structure refinement was calculated according to the Maud Rietveld software to calculate the degree of crystallinity, grain size, and the spacing of crystal faces:

$$CrI = \frac{A_{crystal}}{A_{crystal} + A_{amorph}} \times 100\% \quad (9)$$

where CrI is crystallinity of cellulose, $A_{crystal}$ is calculated area of crystallinity cellulose and A_{amorph} is calculated area of amorphous content.

$$D = \frac{K\lambda}{\beta\cos[\theta]} \quad (10)$$

where K is Scherrer constant, K= 0.9; β is half-width of the diffraction peak; θ is Bragg diffraction angle; λ is X-ray wavelength, which is 1.5419 Å.

$$\lambda = 2dsin[\theta] \quad (11)$$

D represents distance between molecular chains; θ represents angle between incident ray, reflected line and reflecting crystal plane.

2.5.3. Microscopical observation

Characterization of the microscopic surface structure of bamboo powder samples by scanning electron microscopy (SEM) (Regulus 8100, Hitachi, Japan). Samples were gold sprayed before analysis. An accelerating voltage of 10 kV was used during filming. Images were all taken at a magnification of 2000 x. Confocal laser scanning microscopy (CLSM) was used for observation of mosaic bamboo sections. The samples were subjected to autofluorescence detection at an excitation wavelength of 405 nm. Pearson correlation coefficient can be used to measure the linear relationship between variables. The parameters were calculated using Origin software (Version Origin Pro 9.0) according to the following equation.

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} \quad (12)$$

2.5.4. Lignin structural analysis

Extraction of milled wood lignin from raw materials of moso bamboo was carried out using the Bjorkman method. The Soxhlet-extracted bamboo powder was ball-milled for 6 h, then extracted with 96 % dioxane solution (solid–liquid ratio 1:10) for 24 h and the liquid was collected, which was repeated three times. The supernatant was rotary evaporated and centrifuged to obtain crude lignin. The sample obtained was dissolved in 90 % acetic acid solution and then deionized water was added gradually. After centrifugation, precipitate was freeze-dried to obtain crude lignin. The crude lignin was dissolved in dichloroethane/ethanol solution (2:1, v/v) and ether was added gradually, centrifuged, and the lignin was washed with ether and petroleum ether and air-dried to obtain a milled wood lignin sample.

Gradually add 1M HCl solution to the pretreatment black liquor to adjust the pH value and a large amount of lignin precipitated when the pH was adjusted to 6. The lignin was precipitated when the drop was continued until the pH was 2. The lignin samples were centrifuged at 8000 rpm for 10 min, and the lignin precipitates were removed and washed repeatedly with deionized water to neutrality and dried in a vacuum drier at 45 °C overnight. The obtained samples were dissolved in 90 % acetic acid solution and the lignin was precipitated with deionized water. After centrifugation, the lignin was freeze-dried and prepared for use.

Dissolve 50 mg of lignin samples in 0.8 mL of dimethyl sulfoxide

(DMSO)-d6 and perform two-dimensional HSQC NMR analysis. The acquisition time was 256.1 ms with 32 incremental scans, and the acquisition time was 6.1 ms with a 1.5 s delay.

3. Results and discussion

3.1. DES characterization

The viscosity and Kamlet-Taft lysochromic parameters of two different molar ratios of DESs were characterized, and the results are listed in Table 1. Viscosity is an important parameter for DESs as solvents. The viscosities of CA1 and CA2 viscosity at room temperature, are 1026.3 mPa·s and 983.6 mPa·s, respectively. The values were far lower than the other arginine DESs, and even lower than the water content of 6.5 % wt of choline chloride-lactic acid classic system [37,38]. The low viscosity is conducive to the contact between DESs and bamboo feedstock, which may deconstructs lignocellulose through macromolecular mobility in shorter time [39].

The chemical properties of the proposed DESs are reflected by Kamlet-Taft solvent chromism parameters, which include α (hydrogen-bond acidity), β (hydrogen-bond basicity), and π* (polarity/polarizability) values [35]. DES with a larger (β-α) (net alkalinity) value is beneficial for dissolving lignin, while the solubility of xylan is linearly related to the β value [40]. As inferred from Table 1, the π* parameters are 1.270 (CA1) and 1.284 (CA2) respectively, which shows their high polarity and ability to construct non-specific interactions such as dipole–dipole interactions [41]. Meanwhile, CA1 and CA2 exhibit high (β-α) values, indicating their superior lignin removal performance when treating moso bamboo [42]. Higher β values of choline-arginine-based DES suggest a stronger hydrogen bond donor capacity of the DES. Besides, DES with high (β-α) values can form stronger hydrogen bonds with lignin and xylan, facilitating breakage of their non-covalent linkages with cellulose. Therefore, DES with higher β values and (β-α) values are more effective in the pretreatment of moso bamboo biomass [41].

3.2. Chemical compositional analysis

The cellulose recovery and chemical composition of raw and pretreated moso bamboo samples are listed in Fig. 1. The cellulose recovery of bamboo after CA1 pretreatment was generally above 90 %, but it decreased to 86.4 % after CA2 pretreatment. It has been reported that, high lignin removal rates are commonly accompanied by certain amount of sugars loss, which is attributed to the degradation of moso bamboo cellulose under highly alkaline solvents [43]. For example, acidic ChCl/ethylene glycol (EG) DESs were developed and with the optimizing of alkali dosage, the cellulose recovery decreased from 87.43 % to 79.91 % [44]. Therefore, it can be concluded that the lower the β values and (β-α) values, the more favorable the recovery of cellulose after DESs treatment.

When the temperature elevated from 90 °C to 130 °C, the increase of cellulose content and the decrease of lignin and xylan content were also observed, reaching 94.4 % of lignin and 75.6 % of xylan respectively (Fig. 1c and d). The lignin removal rate was significantly improved as compared to ChCl-EG/sodium sulfide DESs treatment by traditional heating method, which removed 56.85–74.67 % lignin [26]. Hence, increasing pretreatment temperature significantly enhanced the removal of lignin and xylan [23]. It was therefore critical to increase the incubation temperature applied to pretreatments on moso bamboo in various solvents treatments.

In addition to pretreatment temperature, previous studies have also shown that pretreatment duration affects lignin removal. Hu et al. suggested that, EG/citric acid (CA) DES remove 47.07 %–51.48 % lignin from bamboo at 90 °C for 4 h–24 h by traditional heating method [45]. The conventional DES system composed of choline chloride/lactic acid (LA) only achieved a lignin removal rate of 84.1 % when treated at the same temperature for 90 min using traditional heating methods [46]. At the same temperature, the microwave-assisted arginine/choline DES pretreatment in this study required only 15 min to obtain a satisfying pretreatment effect. Compared with other arginine-based DESs and amino acid-based DESs, choline-arginine DESs in the present work show evidently better ability to remove lignin and xylan [47,48]. Meanwhile, the application of microwave-assisted pretreatment would make it easier for solvent molecules to penetrate the dense structure of bamboo and contact the binding site more easily, and thus showed stronger lignin and xylan removal ability [49].

Table 1
The Kamlet–Taft solvatochromic parameters of DESs, density and viscosity at room temperature.

DES	Molar ratio	α	β	π*	β-α	Density (g/cm^3)	Viscosity (mPa·s)
CA1	1:1	0.309	0.642	1.270	0.333	1.237	1026.3
CA2	1:2	0.292	0.845	1.284	0.554	1.182	983.6

3.3. Enzymatic hydrolysis

The enzymatic hydrolysis yield is considered as an important index to evaluate the pretreatment effect of DESs. As shown in Fig. 2, direct cellulase hydrolysis of the original sample for 72 h resulted in a rather low glucose yield of 6.2 %, which is consistent with previous research [50] (Fig. 2a). After pretreatment by CA1 and CA2, the glucose yield of bamboo increased to 69.5 %–91.8 % and 72.0 %–86.0 % (Fig. 2b and c), respectively. The glucose yields were 11.2–14.8 and 11.6–13.9 times higher than those of the original samples, respectively. It was higher than the conversion of samples pretreated with guanidine hydrochloride (GH)/urea (U) (90.79 wt%), GH/glycerol (Gly) (86.67 wt%), choline chloride/lactic acid (LA) (72.56 wt%), and GH/LA (50.09 wt%) DESs at 130 °C for 30 min [51]. The removal of lignin-carbohydrate complex by the DESs may be one of the major factors for improving the accessibility of enzymes to cellulose chains. Microwave-assisted CA1 reaction at 130 °C for 15 min can increase the glucose yield to 91.8 %, and CA2 reaction at 90 °C can increase the conversion rate to 86.0 %. The phenomena suggested the ideal pretreatment efficiency by choline-arginine DES properties.

With respect to conversion of hemicelluloses, the yield of xylose increased from 8.9 % to 56.8 %–71.4 % for CA1 and 8.9 % to 68.0 %–76.8 % for CA2 (Fig. 2d, e and f). The results showed that the pretreatment performance of CA1 and CA2 was better than that of most alkaline DESs pretreatments, such as ChCl/guaiacol and ChCl/monoethanolamine DESs [52,53]. It has been reported that, alkaline DESs showed the capability to saponify and split the ester bonds between hemicellulose and other components [50]. In addition to the removal of acetyl groups and the weakening of hydrogen bonds between cellulose and hemicellulose, satisfying enzymatic degradability on both cellulose and hemicelluloses saccharides were achieved [54].

In order to furtherly show the relationship between the lignin and xylan removal rate by DESs and glucose yields by enzymatic hydrolysis, linear fitting plot was performed. As shown in Fig. S1, there was a high linear positive correlation between the removal rate of xylan and lignin of CA1 and its glucose yield, with $R^2 = 0.91$ and 0.99. However, CA2 removal rate was negatively correlated with glucose yield. This is related to the structural changes caused by the removal of the outer envelope structure of cellulose in a highly alkaline environment [55].

3.4. Morphological observation

Morphological changes of bamboo after DESs pretreatment were qualitatively studied by SEM, which is of great significance for explaining the effects of DESs pretreatment on the surface structure and enzymatic hydrolysis of lignocellulose. As seen in Fig. 3, raw material shows long fiber beam structure and irregular aggregate, speculating

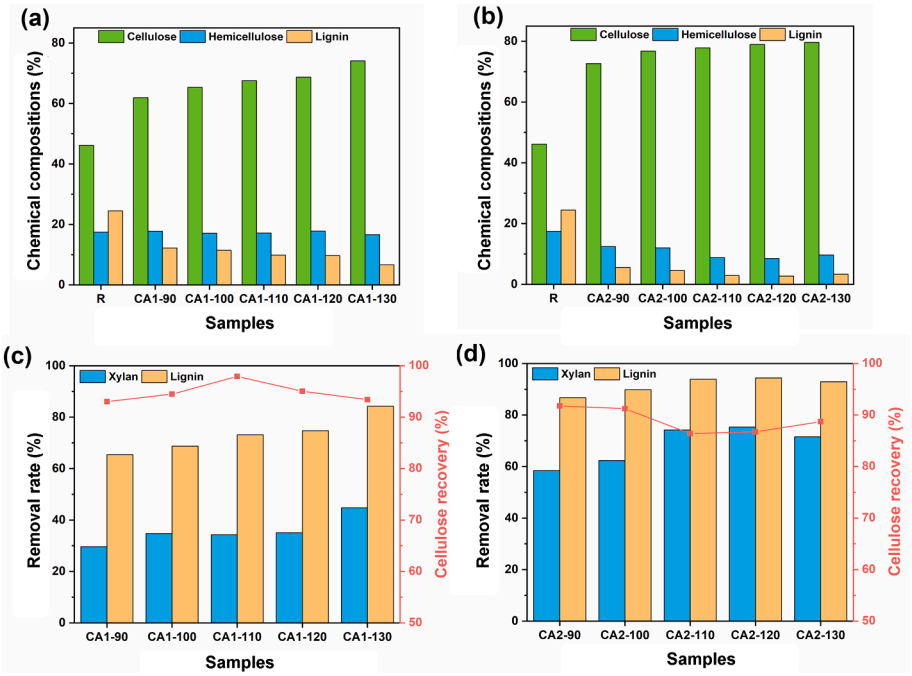


Fig. 1. The chemical composition of samples pretreated by CA1 (a) and CA2 (b) and removal of xylan and delignification of pretreated samples (c) and (d).

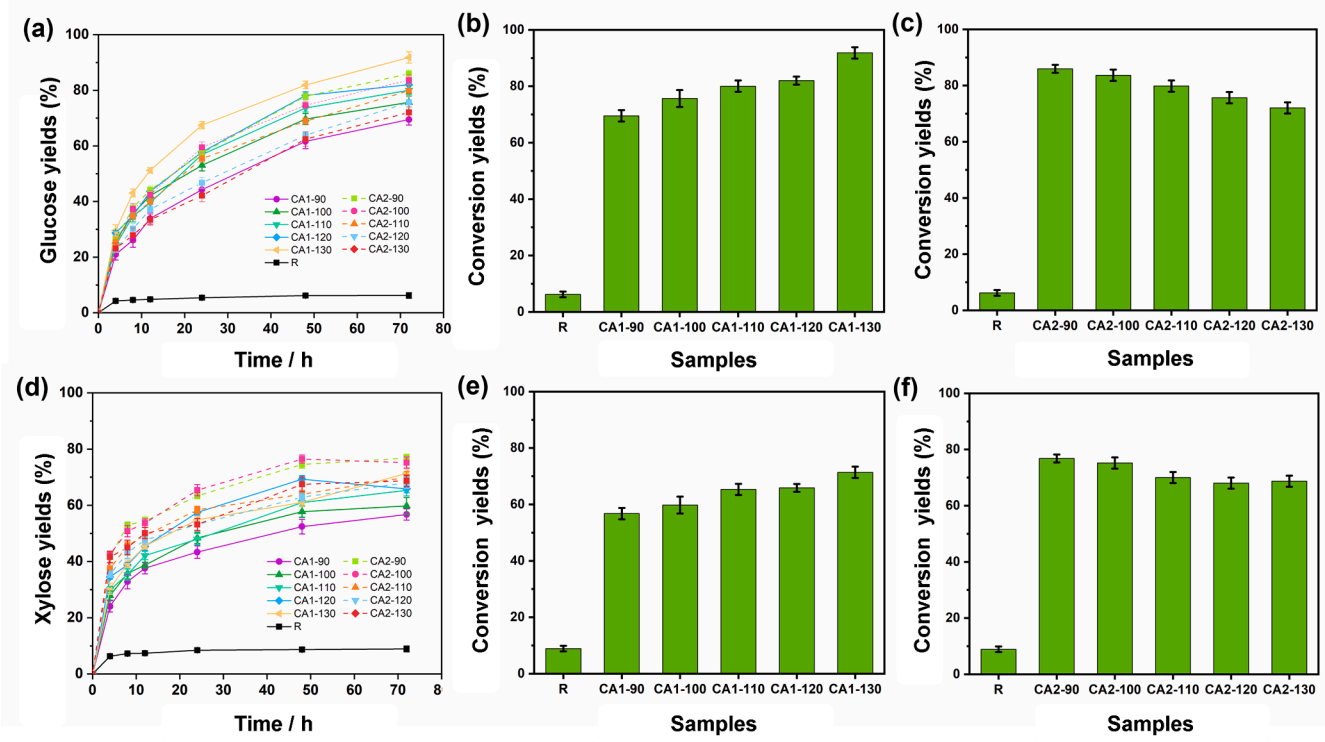


Fig. 2. (a) Glucose yield of samples; (b) 72 h glucose yield of CA1 pretreatment samples; (c) 72 h glucose yield of CA2 pretreatment samples; (d) xylose yield of samples; (e) 72 h xylose yield of CA1 pretreatment samples; (f) 72 h xylose yield of CA2 pretreatment samples.

that the uneven surface structure is due to the structure of the lignin and hemicellulose packages and some residual extract adhesion [56]. Compared with the samples of raw bamboo material and CA1 pretreatment, it was found that the surface of the treated samples showed obvious changes, and the surface roughness and bamboo fiber size were reduced. This is due to the removal of lignin and hemicellulose on the cellulose surface, resulting in the overall surface becoming flat and leaving a dense cellulose structure. This would be favorable for improving the accessibility of bamboo cellulose substrate, and thus increasing the enzymatic hydrolysis yield. The more ordered surface of CA2-treated fibers is also associated with higher lignin-cellulose removal capacity. Even single cellulose fibers can be observed (Fig. 3j and k), which is attributed to the fact that lignin in lignocellulosic biomass acts as a glue to maintain the structure of the cell wall, and the

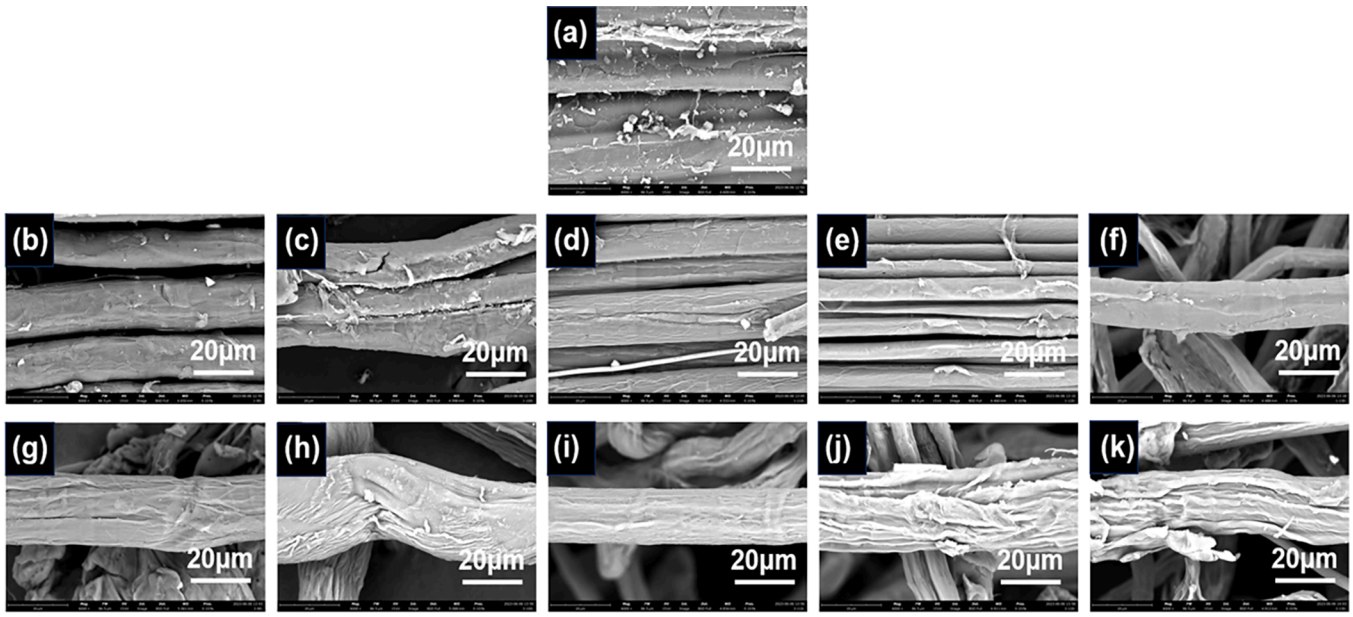


Fig. 3. SEM observations of R (a), CA1-90 (b), CA1-100 (c), CA1-110 (d), CA1-120 (e), CA1-130 (f), CA2-90 (g), CA2-100 (h), CA2-110 (i), CA2-120 (j) and CA2-130 (k).

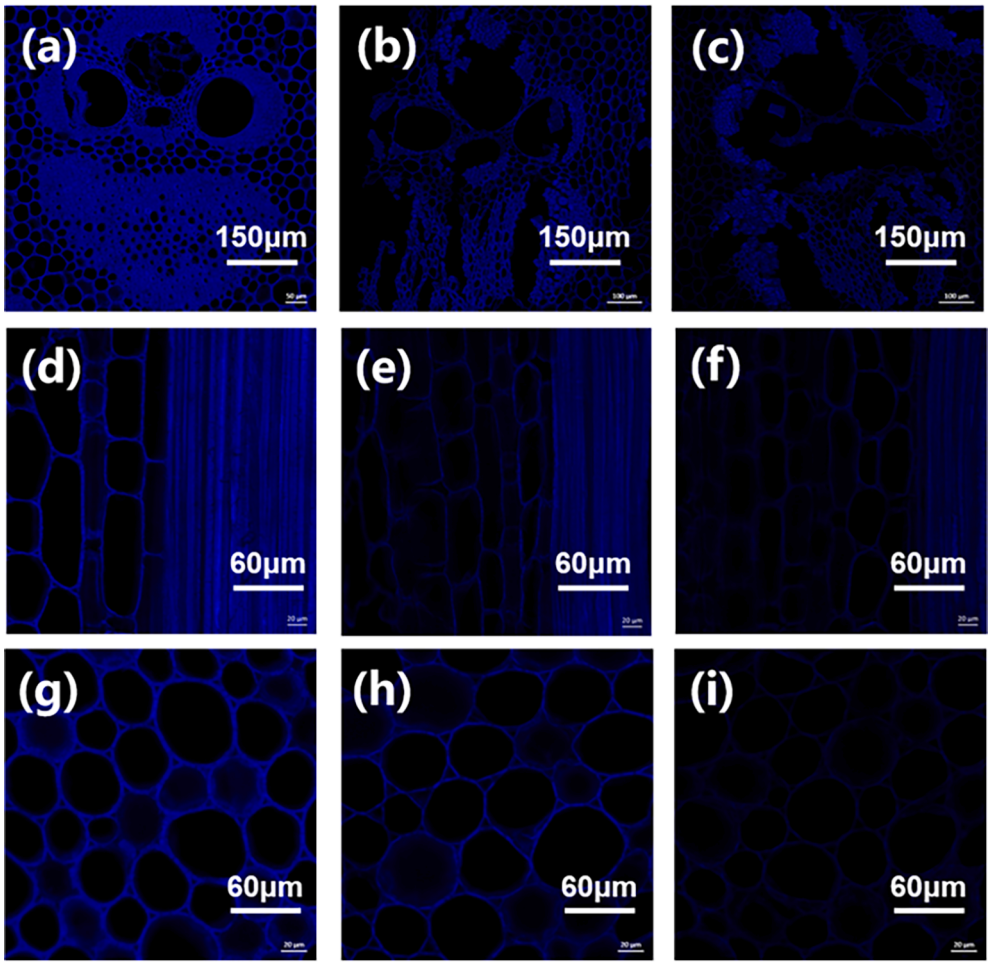


Fig. 4. CLSM micrographs of R-S (a), CA1-90-S (b) and CA2-90-S (c) in transverse section; CLSM micrographs of R-S (d), CA1-90-S (e) and CA2-90-S (f) in radial section and parenchyma cell of R-S (g), CA1-90-S (h), and CA2-90-S (i) in radial section.

destruction of lignin by CA2 leads to the separation of microfibers and adjacent fibers [57]. Besides, alkaline DESs have the property of cellulose swelling as discussed above [58], that is, the entry of solvent molecules into the disordered region of cellulose leads to the weakened hydrogen bond and the increase of molecular chain distance. Therefore, the size of bamboo fiber after CA2 pretreatment shows a trend of increasing, which will provide more effective contact area and active site for cellulase.

CLSM revealed changes in the distribution and content of lignin at the ultrastructural level. As shown in Fig. 4, the autofluorescence method of lignin was employed to observe morphological changes and lignin distribution in different cross-sectional tissues and various cell types of moso bamboo before and after pretreatment [59]. The pristine samples exhibited bright fluorescent intensity, majorly due to the autofluorescence by high amount of lignin in untreated samples. Fluorescent intensities of CA1-90-S and CA2-90-S were significantly reduced as compared to raw materials both on cross sections and longitudinal sections. It furtherly confirmed above component analysis that abundant lignin removal took place within the bamboo cell walls and surface of cellulose fibers. Slices treated by CA2 were visibly darker under CLSM observations. It suggests that, the high lignin removal capacity of CA2 results in the virtual disappearance of the fluorescence signal from the samples, which would also expose the cellulose-accessible surface to facilitate cellulase hydrolysis.

3.5. Crystal structural analysis

The effect of choline-arginine DESs on the crystal structure of bamboo cellulose was analysed by XRD. The XRD spectra of the raw and DESs-treated samples showed similar characteristic peaks (Fig. 5a), with four diffraction peaks at 2θ = 14.9°, 16.7°, 22.2° and 34.8°, which corresponded to (1 −1 0), (1 1 0), (2 0 0) and (0 0 4) surfaces, respectively [60]. The present diffraction peaks are consistent with the standard characteristic peaks of natural cellulose I_β, indicating that the proposed choline-arginine DESs well maintained the initial crystalline allomorphs of bamboo cellulose.

The crystallinity, crystallite size and d-Spacing of the sample obtained after fitting are shown in Fig. 5b and c. The crystallinity of the untreated bamboo sample is about 73.1 %, which increased to 73.5 %–79.5 % after CA1 treatment. The increase of crystallinity was majorly caused by the successful removal of amorphous lignin and hemicelluloses by the DESs. Overall, the crystallinity increases with increasing temperature and decreases slightly as the temperature is further increased above 120 °C due to disintegration of the cellulose crystalline zone. However, as the temperature continues to increase, the amorphous cellulose structure is partially hydrolyzed at 130 °C, which resulted in the higher crystallinity. In terms of the crystallinity variations of cellulose after CA2 treatment, it is interesting to show the decreased values from 73.1 % to 69.2 %–72.3 %, despite the removal of amorphous lignin/hemicellulose components. It is probably due to the solvent CA2 with higher choline ratios and alkalinity, which tended to loosen highly ordered crystalline regions of cellulose chains and thus accelerated their transformation into amorphous parts with highly disordered arrangements [61].

For the followed hydrolysis of pretreated bamboo cellulose by cellulase enzymes, it was reported that, Van der Waals interaction and hydrophobic interaction between tyrosine residues and glucopyranose rings in the binding domain of cellulase were used to adsorb cellulose substrates [62]. In order to evaluate the cellulase binding to the hydrophobic surface of treated cellulose, the typical crystal structure of hydrophobic (2 0 0) plane was quantitatively determined. For raw bamboo cellulose, the crystallite size perpendicular to hydrophobic (2 0 0) plane appeared as 3.4 nm, while CA1 or CA2 pretreated samples showed visibly decreased crystallite sizes to 2.9–3.4 nm and 2.3–2.5 nm

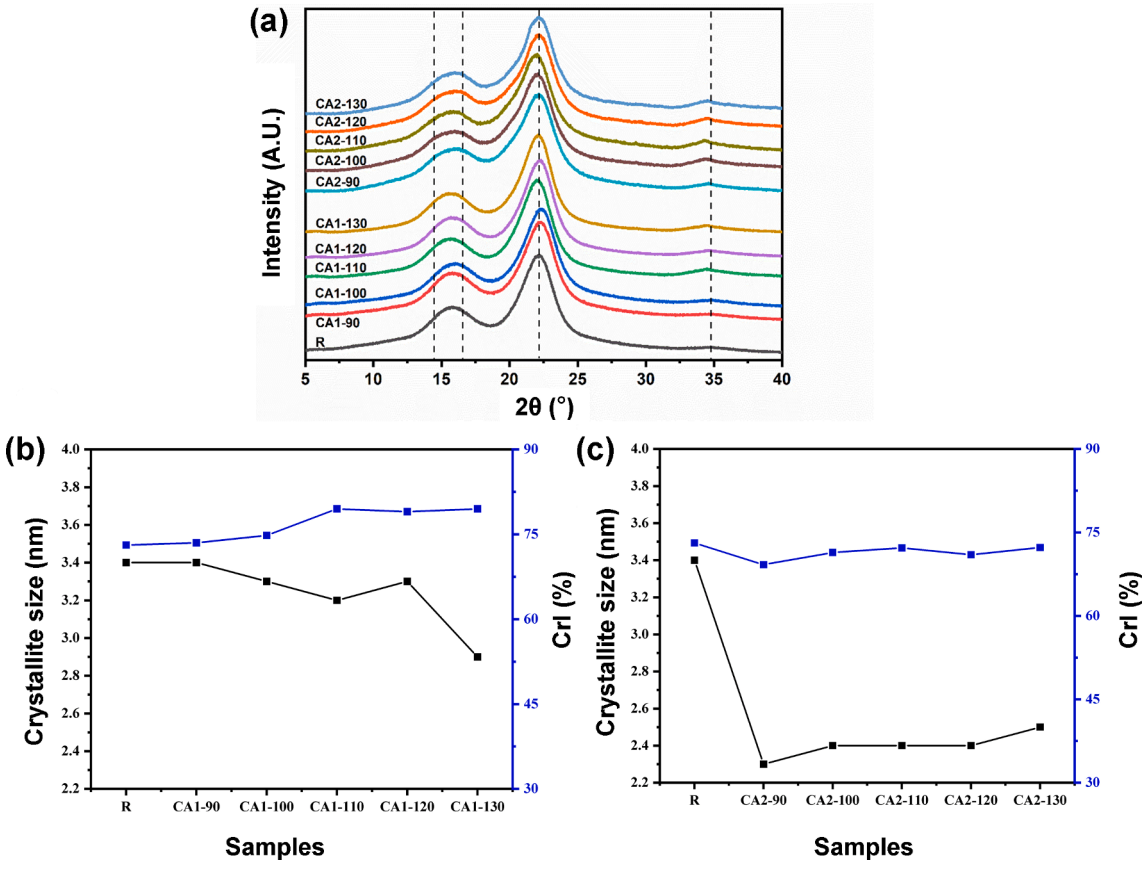


Fig. 5. XRD patterns of pretreated samples (a), crystallite sizes perpendicular to (2 0 0) lattice plane and crystallinity for samples CA1 (b) and CA2 (c).

respectively. The phenomenon of hydrophobic crystallite sizes drop was consistent with the previous results on pretreated cellulosic samples under alkali environments, which may provide more adsorption sites for cellulase and favor the enzymatic hydrolysis yields [63,64].

Based on above investigation on the arginine/choline pretreatment on bamboo cellulose, a 36-chain cellulose elementary fiber (CEF) model based on cellulose I_β and a diagram were proposed to explain inner mechanisms on crystal structural evolution (Fig. 6a). After DESs treatment, the change of spacing between neighboring crystal planes of the cellulose is very limited, while the crystallite sizes showed remarkable reduce, which suggested the lowered cellulose chains remained after the treatments. The successful removal of amorphous lignin and hemicelluloses linked surrounding glucan chain bundles may cause the swelling of the cellulose crystals, especially in alkaline treatment conditions. This behavior furtherly facilitates the transformation of surface chains of cellulose crystals into disordered parts, which took place during the DESs treatments. Thus, fragmentation of the crystal structure into smaller sizes was achieved, favoring the followed enzymatic adsorption and degradation [65].

Fig. 6b and 6c show the correlation of crystalline structure of cellulose with the dissociation behavior of lignin/ xylan and the conversion rate of enzymatic hydrolysis. The removal of lignin and xylan by CA1 had a strong negative correlation with the grain cellulose crystallite sizes ($R^2 = 0.925$ and 0.882). Therefore, higher removal degree of lignin and xylan resulted in the smaller crystallite sizes of cellulose, corresponding to above hypothesis. The crystallite size of CA1 pretreated bamboo cellulose showed a strong linear negative correlation with its enzymatic hydrolysis yield ($R^2 = 0.852$), while the R^2 of CA2 was 0.749. Comparing the effect of crystallinity and crystallite size on the conversion efficiency of cellulase, it was found that the decreased crystallite sizes by CA1 and CA2 treatments had a critical impact on promoting the enzymatic hydrolysis yield.

3.6. Lignin characteristics

2D-HSQC NMR spectroscopy was used to investigate the chemical structural variations of lignin after DESs pretreatment. The two-dimensional map information of aliphatic region and aromatic region is shown in Fig. 7 and annotated according to previous literature [66]. Typical bonds of lignin were present in all samples, including β-O-4 bond structure (A), β-β bond structure (B), and β-5 bond structure (C). Moreover, The observation of γ-acetylated β-O-4 bond (A') signals in pristine lignin samples represents the acetylation of the γ-C site of lignin and the appearance of acylated p-coumaric acid structure [8]. However, the signal was weakened under CA2 treatment at 90 °C and disappeared after CA1 treatment at 130 °C, indicating that the acetyl group was shed under alkaline DESs pretreatment and eliminated at high temperature [67]. At the same time, A_α, A_β (S), and A_β (G) correspond to C_α-H_α, C_β-H_β in β-O-4 bond of eugenic-based lignin and C_β-H_β in β-O-4 bond of guaiac-based lignin, respectively. After DESs treatment, the β-O-4 signal of lignin tends to decrease. It suggests that, alkaline DESs selectively cleaved a large number of β-O-4 ether bonds [4]. In addition to the cleavage of the β-O-4 bond, the contents of retinol β-β structure (B) and phenyl coumarin β-5 structure (C) were also reduced, suggesting that some C-C bonds were also degraded by choline-arginine DESs [68].

The main aromatic unit structure of phyllostachy lignin also appeared in Fig. 7, including syringyl (S), guaiacyl (G), and p-hydroxyphenyl (H). The S/G values of raw bamboo lignin, CA1 and CA2 pretreatment solution lignin were 1.1, 1.4 and 1.9, respectively, which were similar to the S/G results in other studies [69]. G lignin was reported to adsorb cellulase more readily than S lignin and H lignin. Therefore, the enzymatic efficiency of the samples containing more G lignin was lower than that of the samples containing less G lignin and the same trend was observed for moso bamboo [70,71]. Meanwhile, p-coumaric acid (PCA) and ferulic acid (FA) were abundant in the CA2 pretreatment solution, and the obvious signals were located at ($PCA_{2,6}$), ($PCA_{3,5}$), (PCA_β), and (FA_2), respectively. Besides, the α signals of p-coumaric acid and ferulic acid overlapped at δC/δH144.6/7.5 ppm. It was seen that, FA and PCA are cleaved after treatment with the alkaline deep eutectic solvent CA2, preventing the formation of ester or ether bonds between the lignin and hemicelluloses. Therefore, the recalcitrance of the substrates caused by lignin-carbohydrate complex was proved to be cleaved by the proposed treatments, which enhanced the contact between substrates and cellulase enzymes [72].

3.7. Mass balance

Mass balance of the samples was performed based on 100 g poplar trees to evaluate the efficiency of choline-arginine DES pretreatment, as shown in Fig. 8. Treatment condition of 130 °C for 15 min by microwave-assisted CA1 incubation was selected for the balance calculation. After the treatment, solid residue that was majorly composed of

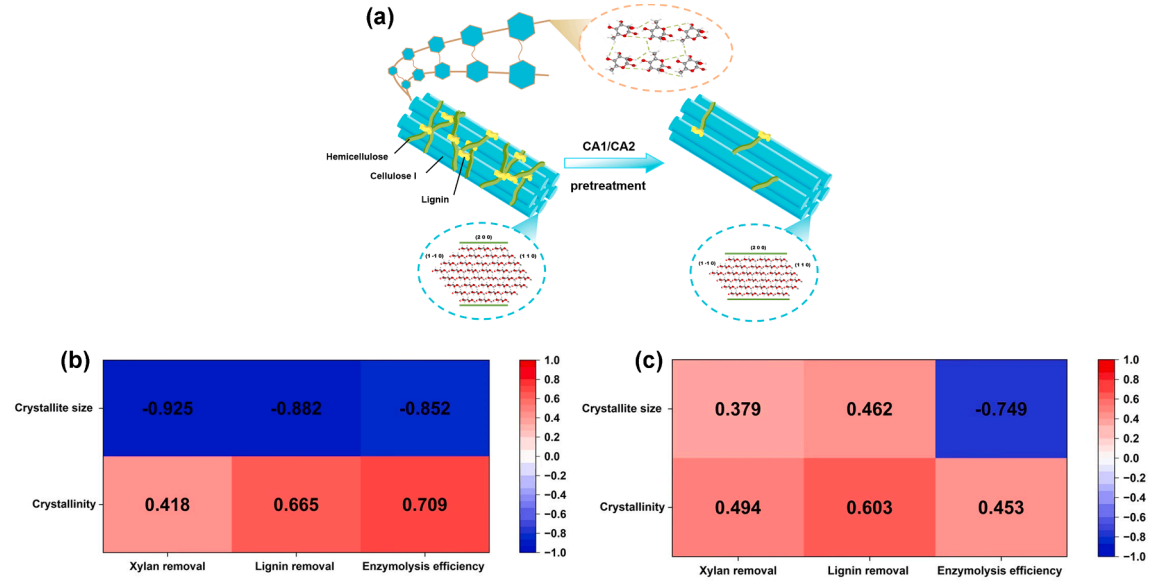


Fig. 6. Schematic model for cellulose crystalline structural variations during DESs pretreatment (a); the correlation of crystallite size and crystallinity with hemicellulose removal, lignin removal and enzymatic hydrolysis yield of samples pretreated by (b) CA1and (c) CA2.

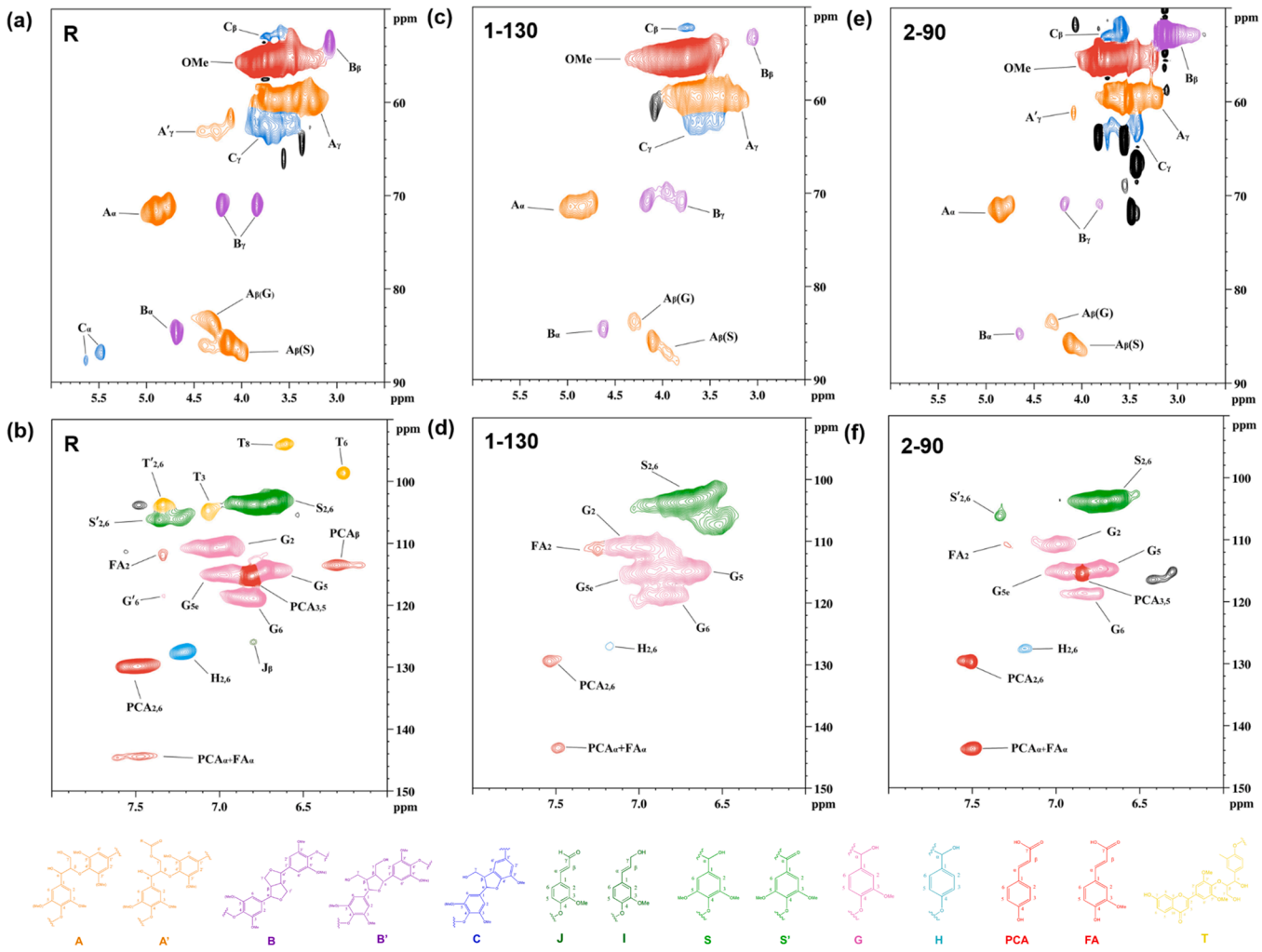


Fig. 7. The 2D HSQC NMR spectra and key subunits & interunit linkages of lignin.

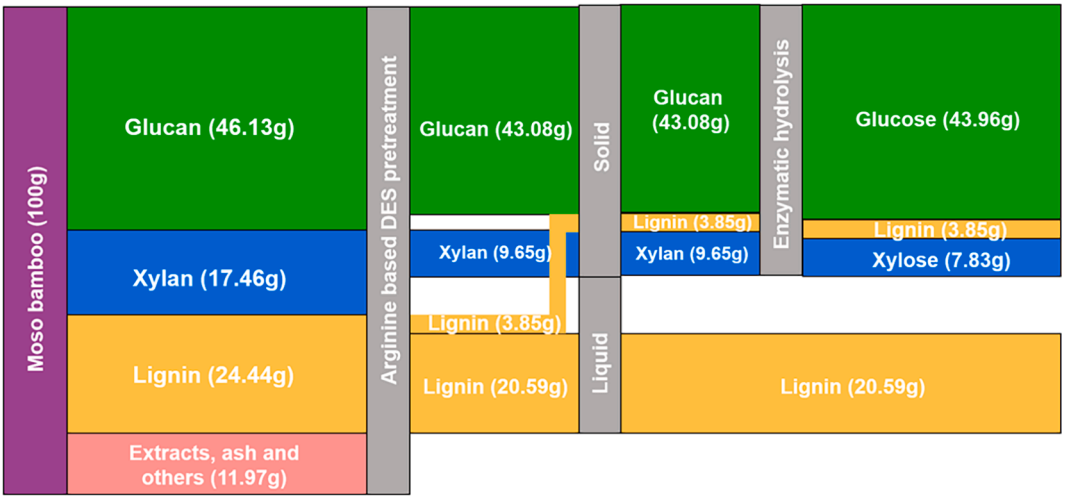


Fig. 8. Material balance in choline-arginine DESs pretreatment (sample treated by CA1 at 130 °C for 15 min with microwave assistance as example).

cellulose and part of xylan are easily separated from the pretreated mixture. The solid residue obtained weighed 58.16 g and, based on component analysis, consisted of 43.08 g glucan, 9.65 g xylan, and 3.85 g lignin. Through cellulase hydrolysis with a 5 % substrate concentration and 20 FPU/g enzyme for 72 h, a total of 51.79 g fermentable sugars was obtained, including 43.96 g glucose and 7.83 g xylose. This was significantly higher than the enzymatic hydrolysis yield of ChCl-EG-Na_2S-based alkaline ternary DESs after pretreating moso bamboo at 140 °C for 2 h (the glucose yield was 42.66 g after 72 h reaction with 25 FPU/g enzyme dosage) [24]. Similarly, this result was higher than the

conversion of poplar pomace (31.60 g of glucose yield after 72 h of cellulase hydrolysis) obtained after a two-step pretreatment with hydrothermal and microwave-assisted DESs (hydrothermal treatment stage: 140 °C, 0.01 M $AlCl_3$, 1 h; DESs treatment stage: ChCl-EG-$AlCl_3$ after 100 °C, 10 min) [73]. In addition, many lignin structures rich in β-O-4 bonds can be obtained in the liquid. At the same time, the pretreated DESs can also be reused after recycling, which can effectively reduce the cost of pretreatment and make the process renewable and sustainable.

4. Conclusion

This study developed a rapid microwave-assisted choline-arginine deep eutectic solvent (DES) pretreatment to optimize the fractionation and enzymatic conversion of moso bamboo lignocellulose, revealing its high potential for selective delignification, carbohydrate recovery, and fermentable sugar production. Its high β value (0.845) and (β-α) value (0.554) demonstrate high selectivity for delignification, excellent carbohydrate recovery, and desirable fermentable sugar yields. Under optimal conditions (130 °C, 15 min, arginine:ChOH = 1:1, 20 FPU/g), it achieved a high delignification rate (94.4 %) and xylan removal rate (75.6 %), resulting in enhanced dual monosaccharide yields, with 91.8 % glucose yield and 71.4 % xylose yield. Moreover, in the alkaline environment, the degradation of polysaccharides was greatly minimized, and the cleavage of alkali-unstable FA and PCA by choline-arginine DESs was observed, which were considered favorable factors for improving enzymatic hydrolysis yields. Pretreatment conditions under selected microwave-assisted methods were more effective in both improving results and reducing energy consumption compared to conventional heating conditions. Overall, the proposed arginine/ChOH DESs pretreatment strategy proved to be a low-cost and effective method, showing great application potential for lignocellulose deconstruction and conversion into bio-based chemicals.

CRediT authorship contribution statement

Junpeng Liu: Writing – original draft, Visualization, Formal analysis. **Ying Xie:** Validation, Data curation. **Xuan Xu:** Investigation. **Rongrong Xiang:** Methodology. **Chengyu Zhang:** Resources. **Zhe Ji:** Writing – review & editing, Funding acquisition, Conceptualization. **Qiang Yong:** Supervision, Project administration. **Zhe Ling:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2025.160909.

Data availability

Data will be made available on request.

References

[1] J. Cheng, C. Huang, Y. Zhan, S. Han, J. Wang, X. Meng, C.G. Yoo, G. Fang, A. J. Ragauskas, Effective biomass fractionation and lignin stabilization using a diol DES system, Chem. Eng. J. 443 (2022) 136395.
[2] V. Sharma, M.-L. Tsai, C.-W. Chen, P.-P. Sun, A.K. Patel, R.R. Singhania, P. Nargotra, C.-D. Dong, Deep eutectic solvents as promising pretreatment agents for sustainable lignocellulosic biorefineries: a review, Bioresour. Technol. 360 (2022) 127631.
[3] C.-Y. Ma, L.-H. Xu, Q. Sun, X.-J. Shen, J.-L. Wen, T.-Q. Yuan, Tailored one-pot lignocellulose fractionation to maximize biorefinery toward controllable producing lignin nanoparticles and facilitating enzymatic hydrolysis, Chem. Eng. J. 450 (2022) 138315.
[4] X.-J. Shen, T. Chen, H.-M. Wang, Q. Mei, F. Yue, S. Sun, J.-L. Wen, T.-Q. Yuan, R.-C. Sun, Structural and morphological transformations of lignin macromolecules during bio-based deep eutectic solvent (DES) pretreatment, ACS Sustain. Chem. Eng. 8 (5) (2019) 2130–2137.
[5] W. Ren, D. Zhang, Y. Zhou, H. Wang, L. Xia, C. Tan, F. Guo, X. Zhang, R. Yang, Y. Yu, Lignin's complex role in lignocellulosic biomass Recalcitrance: a case study on bamboo, Chem. Eng. J. 490 (2024) 151422.
[6] Y. Ci, Y. Ma, T. Chen, F. Li, Y. Tang, Facile dissolution of cellulose by superbase-derived ionic liquid using organic solvents as co-solvents at mild temperatures, Carbohydr. Polym. 330 (2024) 121836.
[7] P. Halder, S. Kundu, S. Patel, A. Setiawan, R. Atkin, R. Parthasarthy, J. Paz-Ferreiro, A. Surapaneni, K. Shah, Progress on the pre-treatment of lignocellulosic biomass employing ionic liquids, Renew. Sustain. Energy Rev. 105 (2019) 268–292.
[8] W.-X. Li, W.-Z. Xiao, Y.-Q. Yang, Q. Wang, X. Chen, L.-P. Xiao, R.-C. Sun, Insights into bamboo delignification with acidic deep eutectic solvents pretreatment for enhanced lignin fractionation and valorization, Ind. Crop. Prod. 170 (2021) 113692.
[9] Y. Ci, D. Lv, X. Yang, H. Du, Y. Tang, High-performance cellulose/thermoplastic polyurethane composites enabled by interaction-modulated cellulose regeneration, Carbohydr. Polym. 346 (2024) 122611.
[10] X. Dong, X. Yang, Y. Ci, F. Li, T. Chen, Y. Li, Y. Tang, Preparation and characterization of cellulose phosphite ester with high flame retardancy via transesterification in BmimCl/DMSO system, Ind. Crop. Prod. 216 (2024) 118694.
[11] Y. Zhong, J. Wu, H. Kang, R. Liu, Choline hydroxide based deep eutectic solvent for dissolving cellulose, Green Chem. 24 (6) (2022) 2464–2475.
[12] T. Xiao, M. Hou, X. Guo, X. Cao, C. Li, Q. Zhang, W. Jia, Y. Sun, Y. Guo, H. Shi, Recent progress in deep eutectic solvent (DES) fractionation of lignocellulosic components: a review, Renew. Sustain. Energy Rev. 192 (2024) 114243.
[13] Y.-L. Chen, X. Zhang, T.-T. You, F. Xu, Deep eutectic solvents (DESs) for cellulose dissolution: a mini-review, Cellul. 26 (2019) 205–213.
[14] Z. Tang, C. Wu, W. Tang, C. Ma, Y.-C. He, A novel cetyltrimethylammonium bromide-based deep eutectic solvent pretreatment of rice husk to efficiently enhance its enzymatic hydrolysis, Bioresour. Technol. 376 (2023) 128806.
[15] S. Tang, Q. Dong, Z. Fang, Z.-D. Miao, Complete recovery of cellulose from rice straw pretreated with ethylene glycol and aluminum chloride for enzymatic hydrolysis, Bioresour. Technol. 284 (2019) 98–104.
[16] W. He, D. Yang, W. Tang, C. Ma, Y.-C. He, Integrated understanding of enhancing enzymatic hydrolysis of bulrush through the metal chloride-assisting choline chloride: glycerol pretreatment, Ind. Crop. Prod. 205 (2023) 117586.
[17] Z.-K. Wang, S. Hong, J.-L. Wen, C.-Y. Ma, L. Tang, H. Jiang, J.-J. Chen, S. Li, X.-J. Shen, T.-Q. Yuan, Lewis acid-facilitated deep eutectic solvent (DES) pretreatment for producing high-purity and antioxidative lignin, ACS Sustain. Chem. Eng. 8 (2) (2019) 1050–1057.
[18] Y. Bai, X.-F. Zhang, Z. Wang, T. Zheng, J. Yao, Deep eutectic solvent with bifunctional Brønsted-Lewis acids for highly efficient lignocellulose fractionation, Bioresour. Technol. 347 (2022) 126723.
[19] K.H. Kim, T. Dutta, J. Sun, B. Simmons, S. Singh, Biomass pretreatment using deep eutectic solvents from lignin derived phenols, Green Chem. 20 (4) (2018) 809–815.
[20] Z. Chen, Y. Wang, H. Cheng, H. Zhou, Hemicellulose degradation: an overlooked issue in acidic deep eutectic solvents pretreatment of lignocellulosic biomass, Ind. Crop. Prod. 187 (2022) 115335.
[21] K. Wang, H. Yang, X. Yao, F. Xu, R.-C. Sun, Structural transformation of hemicelluloses and lignin from triploid poplar during acid-pretreatment based biorefinery process, Bioresour. Technol. 116 (2012) 99–106.
[22] Y. Guo, L. Xu, F. Shen, J. Hu, M. Huang, J. He, Y. Zhang, S. Deng, Q. Li, D. Tian, Insights into lignocellulosic waste fractionation for lignin nanospheres fabrication using acidic/alkaline deep eutectic solvents, Chemosphere 286 (2022) 131798.
[23] J. Yang, Y. Wang, W. Zhang, M. Li, F. Peng, J. Bian, Alkaline deep eutectic solvents as novel and effective pretreatment media for hemicellulose dissociation and enzymatic hydrolysis enhancement, Int. J. Biol. Macromol. 193 (2021) 1610–1616.
[24] P. Binod, R. Sindhu, R.R. Singhania, S. Vikram, L. Devi, S. Nagalakshmi, N. Kurien, R.K. Sukumaran, A. Pandey, Bioethanol production from rice straw: an overview, Bioresour. Technol. 101 (13) (2010) 4767–4774.
[25] Y. Sheng, S.S. Lam, Y. Wu, S. Ge, J. Wu, L. Cai, Z. Huang, Q. Van Le, C. Sonne, C. Xia, Enzymatic conversion of pretreated lignocellulosic biomass: a review on influence of structural changes of lignin, Bioresour. Technol. 324 (2021) 124631.
[26] X. Liu, J. Cheng, C. Huang, J. Wang, G. Fang, K. Shen, X. Meng, A.J. Ragauskas, Alkali-facilitated deep eutectic solvent for effective bamboo saccharification, Bioresour. Technol. 367 (2023) 128297.
[27] S. Arora, N. Gupta, V. Singh, Choline based basic ionic liquid (BIL)/acidic DES mediated cellulose rich fractionation of agricultural waste biomass and valorization to 5-HMF, Waste Biomass Valoriz. 11 (2020) 3345–3354.

[28] N. Pan, B. Jiang, J. Hu, M. Huang, J. He, Q. Jiang, L. Zhao, F. Shen, D. Tian, The coupling effects between acid-catalyzed hydrothermal pretreatment and acidic/alkaline deep eutectic solvent extraction for wheat straw fractionation, Bioresour. Technol. 386 (2023) 129579.
[29] C. Alvarez-Vasco, R. Ma, M. Quintero, M. Guo, S. Geleynse, K.K. Ramasamy, M. Wolcott, X. Zhang, Unique low-molecular-weight lignin with high purity extracted from wood by deep eutectic solvents (DES): a source of lignin for valorization, Green Chem. 18 (19) (2016) 5133–5141.
[30] M.S. Rahman, R. Roy, B. Jadhav, M.N. Hossain, M.A. Halim, D.E. Raynie, Formulation, structure, and applications of therapeutic and amino acid-based deep eutectic solvents: an overview, J. Mol. Liq. 321 (2021) 114745.
[31] C. Zhang, X. Ma, Y. Gu, Amino acids-based deep eutectic solvents as additives for improved enantioseparation in capillary electrophoresis, J. Sep. Sci. 47 (9–10) (2024) 2400122.
[32] Y. Xie, J. Zhao, P. Wang, Z. Ling, Q. Yong, Natural arginine-based deep eutectic solvents for rapid microwave-assisted pretreatment on crystalline bamboo cellulose with boosting enzymatic conversion performance, Bioresour. Technol. 385 (2023) 129438.
[33] Y. Liang, W. Duan, X. An, Y. Qiao, Y. Tian, H. Zhou, Novel betaine-amino acid based natural deep eutectic solvents for enhancing the enzymatic hydrolysis of corncob, Bioresour. Technol. 310 (2020) 123389.
[34] C. Florindo, A. McIntosh, T. Welton, L. Branco, I. Marrucho, A closer look into deep eutectic solvents: exploring intermolecular interactions using solvatochromic probes, PCCP 20 (1) (2018) 206–213.
[35] X. Liang, Y. Zhu, B. Qi, S. Li, J. Luo, Y. Wan, Structure-property-performance relationships of lactic acid-based deep eutectic solvents with different hydrogen bond acceptors for corn stover pretreatment, Bioresour. Technol. 336 (2021) 125312.
[36] J.K. Kim, J. Yang, S.Y. Park, J.-H. Yu, K.H. Kim, Cellulase recycling in high-solids enzymatic hydrolysis of pretreated empty fruit bunches, Biotechnol. Biofuels 12 (2019) 1–9.
[37] M.S. Rahman, R. Roy, C. Montoya, M.A. Halim, D.E. Raynie, Acidic and basic amino acid-based novel deep eutectic solvents and their role in depolymerization of lignin, J. Mol. Liq. 362 (2022) 119751.
[38] R. Alcalde, A. Gutiérrez, M. Atilhan, S. Aparicio, An experimental and theoretical investigation of the physicochemical properties on choline chloride–Lactic acid based natural deep eutectic solvent (NADES), J. Mol. Liq. 290 (2019) 110916.
[39] E.K. New, T.Y. Wu, C.B.T.L. Lee, Z.Y. Poon, Y.-L. Loow, L.Y.W. Foo, A. Procentese, L.F. Siow, W.H. Teoh, N.N.N. Daud, Potential use of pure and diluted choline chloride-based deep eutectic solvent in delignification of oil palm fronds, Process Saf. Environ. Prot. 123 (2019) 190–198.
[40] L. Zhang, H. Yu, S. Liu, Y. Wang, T. Mu, Z. Xue, Kamlet–taft parameters of deep eutectic solvents and their relationship with dissolution of main lignocellulosic components, Ind. Eng. Chem. Res. 62 (29) (2023) 11723–11734.
[41] Q. Liu, H. Mou, W. Chen, X. Zhao, H. Yu, Z. Xue, T. Mu, Highly efficient dissolution of lignin by eutectic molecular liquids, Ind. Eng. Chem. Res. 58 (51) (2019) 23438–23444.
[42] S. Hong, X.-J. Shen, B. Pang, Z. Xue, X.-F. Cao, J.-L. Wen, Z.-H. Sun, S.S. Lam, T.-Q. Yuan, R.-C. Sun, In-depth interpretation of the structural changes of lignin and formation of diketones during acidic deep eutectic solvent pretreatment, Green Chem. 22 (6) (2020) 1851–1858.
[43] J. Xie, Z. Cheng, S. Zhu, J. Xu, Lewis base enhanced neutral deep eutectic solvent pretreatment for enzymatic hydrolysis of corn straw and lignin characterization, Renew. Energy 188 (2022) 320–328.
[44] H. Li, X. Li, D. Li, J. Zhang, H. Nawaz, T. You, F. Xu, Highly-efficient pretreatment using alkaline enhanced aqueous deep eutectic solvent to unlock poplar for high yield of fermentable sugars: Synergistic removal of lignin and mannan, Bioresour. Technol. 351 (2022) 126993.
[45] Q. Hu, Y. Xu, Y. Wang, W. Gong, C.-Y. Ma, S. Li, J.-L. Wen, Promoting the disassemble and enzymatic saccharification of bamboo shoot shells via efficient hydrated alkaline deep eutectic solvent pretreatment, Int. J. Biol. Macromol. 264 (2024) 130702.
[46] W. Lin, S. Xing, Y. Jin, X. Lu, C. Huang, Q. Yong, Insight into understanding the performance of deep eutectic solvent pretreatment on improving enzymatic digestibility of bamboo residues, Bioresour. Technol. 306 (2020) 123163.
[47] Y. Wang, W.-J. Zhang, J.-Y. Yang, M.-F. Li, F. Peng, J. Bian, Efficient fractionation of woody biomass hemicelluloses using cholinium amino acids-based deep eutectic solvents and their aqueous mixtures, Bioresour. Technol. 354 (2022) 127139.
[48] B.L. Gadilohar, G.S. Shankarling, Choline based ionic liquids and their applications in organic transformation, J. Mol. Liq. 227 (2017) 234–261.
[49] C.-Y. Ma, L.-H. Xu, Q. Sun, S.-N. Sun, X.-F. Cao, J.-L. Wen, T.-Q. Yuan, Ultrafast alkaline deep eutectic solvent pretreatment for enhancing enzymatic saccharification and lignin fractionation from industrial xylose residue, Bioresour. Technol. 352 (2022) 127065.
[50] Z. Guo, J. Mao, Q. Zhang, F. Xu, Integrated biorefinery of bamboo for fermentable sugars, native-like lignin, and furfural production by novel deep eutectic solvents treatment, Ind. Crop. Prod. 188 (2022) 115453.
[51] G. Grillo, E. Calcio Gaudino, R. Rosa, C. Leonelli, A. Timonina, S. Grygiškis, S. Tabasso, G. Cravotto, Green deep eutectic solvents for microwave-assisted biomass delignification and valorisation, Molecules 26 (4) (2021) 798.
[52] Z. Zhao, X. Chen, M.F. Ali, A.A. Abdeltawab, S.M. Yakout, G. Yu, Pretreatment of wheat straw using basic ethanolamine-based deep eutectic solvents for improving enzymatic hydrolysis, Bioresour. Technol. 263 (2018) 325–333.
[53] C. Huang, Y. Zhan, J. Cheng, J. Wang, X. Meng, X. Zhou, G. Fang, A.J. Ragauskas, Facilitating enzymatic hydrolysis with a novel guaiacol-based deep eutectic solvent pretreatment, Bioresour. Technol. 326 (2021) 124696.
[54] W. Wang, T. Yuan, B. Cui, Y. Dai, Investigating lignin and hemicellulose in white rot fungus-pretreated wood that affect enzymatic hydrolysis, Bioresour. Technol. 134 (2013) 381–385.
[55] F. Huang, Z. Tian, H. Ma, Z. Ding, X. Ji, C. Si, D. Wang, Combined alkali impregnation and poly dimethyl diallyl ammonium chloride-assisted cellulase absorption for high-efficiency pretreatment of wheat straw, Adv. Compos. Hybrid Mater. 6 (6) (2023) 230.
[56] X. Qi, J. Chu, L. Jia, A. Kumar, Influence of different pretreatments on the structure and hydrolysis behavior of bamboo: a comparative study, Materials 12 (16) (2019) 2570.
[57] J.A. Sirviö, K. Hyypiö, S. Asaadi, K. Junka, H. Liimatainen, High-strength cellulose nanofibers produced via swelling pretreatment based on a choline chloride–imidazole deep eutectic solvent, Green Chem. 22 (5) (2020) 1763–1775.
[58] Z.X. Zhang XiMing, Q.T. Qu TianJiao, N. Mosier, H.L. Han LuJia, X.W. Xiao WeiHua, Cellulose modification by recyclable swelling solvents, (2018).
[59] X. Wang, Y. Liu, J. Pu, C. Qin, S. Yao, S. Wang, C. Liang, A comparative study on the structure of lignin-carbohydrate complexes in alkali-soluble hemicellulose from bamboo (Bambusa chungii) fibers and parenchyma cells, Ind. Crop. Prod. 210 (2024) 118061.
[60] J. Zhang, Y. Wang, L. Zhang, R. Zhang, G. Liu, G. Cheng, Understanding changes in cellulose crystalline structure of lignocellulosic biomass during ionic liquid pretreatment by XRD, Bioresour. Technol. 151 (2014) 402–405.
[61] P. Tao, Y. Zhang, Z. Wu, X. Liao, S. Nie, Enzymatic pretreatment for cellulose nanofibrils isolation from bagasse pulp: transition of cellulose crystal structure, Carbohydr. Polym. 214 (2019) 1–7.
[62] F. Jiang, J.D. Kittle, X. Tan, A.R. Esker, M. Roman, Effects of sulfate groups on the adsorption and activity of cellulases on cellulose substrates, Langmuir 29 (10) (2013) 3280–3291.
[63] A. Mittal, R. Katahira, M.E. Himmel, D.K. Johnson, Effects of alkaline or liquid-ammonia treatment on crystalline cellulose: changes in crystalline structure and effects on enzymatic digestibility, Biotechnol. Biofuels 4 (2011) 1–16.
[64] Z. Ling, S. Chen, X. Zhang, F. Xu, Exploring crystalline-structural variations of cellulose during alkaline pretreatment for enhanced enzymatic hydrolysis, Bioresour. Technol. 224 (2017) 611–617.
[65] M. Deraman, S. Zakaria, J.A. Murshidi, Estimation of crystallinity and crystallite size of cellulose in benzylated fibres of oil palm empty fruit bunches by X-ray diffraction, Jpn. J. Appl. Phys. 40 (5R) (2001) 3311.
[66] Y. Zhu, T.-X. Yang, B.-K. Qi, H. Li, Q.-S. Zhao, B. Zhao, Acidic and alkaline deep eutectic solvents (DESs) pretreatment of grapevine: component analysis, characterization, lignin structural analysis, and antioxidant properties, Int. J. Biol. Macromol. 236 (2023) 123977.
[67] L. Tan, W. Sun, X. Li, J. Zhao, Y. Qu, Y.M. Choo, S.K. Loh, Bisulfite pretreatment changes the structure and properties of oil palm empty fruit bunch to improve enzymatic hydrolysis and bioethanol production, Biotechnol. J. 10 (6) (2015) 915–925.
[68] H. Yang, C.G. Yoo, X. Meng, Y. Pu, W. Muchero, G.A. Tuskan, T.J. Tschaplinski, A. J. Ragauskas, L. Yao, Structural changes of lignins in natural Populus variants during different pretreatments, Bioresour. Technol. 295 (2020) 122240.
[69] C. Huang, G. Fang, Y. Zhou, X. Du, L. Yu, X. Meng, M. Li, C.G. Yoo, B. Chen, S. Zhai, Increasing the carbohydrate output of bamboo using a combinatorial pretreatment, ACS Sustain. Chem. Eng. 8 (19) (2020) 7380–7393.
[70] Y. Yuan, B. Jiang, H. Chen, W. Wu, S. Wu, Y. Jin, H. Xiao, Recent advances in understanding the effects of lignin structural characteristics on enzymatic hydrolysis, Biotechnol. Biofuels 14 (2021) 1–20.
[71] W. Lin, D. Ning, D. Xu, S. Wu, Q. Zou, A.K. Mondal, F. Huang, Study on the effect of combined and free lignin to the cellulose enzymatic hydrolysis, Ind. Crop. Prod. 222 (2024) 119516.
[72] N. Kothari, S. Bhagia, Y. Pu, C.G. Yoo, M. Li, S. Venketachalam, S. Pattathil, R. Kumar, C.M. Cai, M.G. Hahn, The effect of switchgrass plant cell wall properties on its deconstruction by thermochemical pretreatments coupled with fungal enzymatic hydrolysis or Clostridium thermocellum consolidated bioprocessing, Green Chem. 22 (22) (2020) 7924–7945.
[73] C.-Y. Ma, L.-H. Xu, C. Zhang, K.-N. Guo, T.-Q. Yuan, J.-L. Wen, A synergistic hydrothermal-deep eutectic solvent (DES) pretreatment for rapid fractionation and targeted valorization of hemicelluloses and cellulose from poplar wood, Bioresour. Technol. 341 (2021) 125828.